

Ekene Masha

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer who designs, ships, and operates production systems end to end. Built and deployed TrainerOS, a live fitness coaching platform on ASP.NET Core 8 and PostgreSQL with an asynchronous Azure Functions worker, and Plant Plotter, a full-stack Next.js and Express application on Azure. Proficient in C#, TypeScript, JavaScript, and Python across REST API development, object-oriented design, and distributed messaging. Experienced in large language model (LLM) output evaluation and data quality workflows, working spec-first with mutation-tested code and automated CI/CD.

TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript (ES6+), Python, SQL, PHP, HTML

Frameworks & Libraries: ASP.NET Core 8 (minimal APIs), Entity Framework Core, React 19, Next.js, Express.js, Flask, Vite, Tailwind CSS, Laravel, WordPress

Back-End & Architecture: REST API development, OOP, system design, distributed messaging, queue-based workers, idempotency and retry design, JWT and HMAC authentication, Argon2id hashing, Agile SDLC

Cloud & DevOps: Azure App Service, Azure Functions (isolated worker), Azure Queue Storage, Application Insights, Azure VM, Vercel, GitHub Actions CI/CD, Docker, Linux, Git

Databases & Testing: PostgreSQL (Neon), MySQL, SQL Server, SQLite, MongoDB familiarity, mutation testing, unit and integration testing, Jest, Postman, WCAG AA accessibility

AI & Machine Learning: LLM evaluation and output validation, prompt engineering, natural language processing (NLP) concepts, responsible AI practices, AI-assisted development (Claude Code, Cursor, GitHub Copilot)

PROFESSIONAL EXPERIENCE

The UPS Store

Ottawa, ON

Web Designer / IT Support

Sept 2024 – Oct 2025

- Reduced recurring website errors by diagnosing frontend and backend defects across database-driven workflows, applying structured debugging and root-cause analysis.
- Improved system reliability by resolving staff-reported issues involving login errors, browser conflicts, and database inconsistencies using object-oriented debugging approaches.
- Introduced pre-deployment testing procedures that validated application changes before production updates, accelerating deployment stability and reducing post-release defects.
- Maintained internal technical records for issue tracking, reducing recurring incidents through documented reproducible bugs and structured QA processes aligned with SDLC best practices.

Invisible Technologies

Remote (Ottawa, ON)

AI Data Trainer

June 2025 – Oct 2025

- Improved LLM evaluation accuracy by identifying and documenting reproducible defects in large language model outputs, applying structured quality standards aligned with responsible AI practices.
- Validated AI-generated text outputs against defined eligibility criteria using natural language processing (NLP) assessment frameworks, increasing validation efficiency and reducing recurring output errors.
- Applied SQL review processes to enforce data quality standards across large evaluation datasets, contributing to measurable reduction in model defects across Agile review iterations.
- Strengthened model debugging workflows by systematically documenting edge cases and defect patterns used to retrain and improve LLM performance across SDLC stages.

UPS

Ottawa, ON

Package Delivery Driver

Oct 2025 – June 2026

- Managed high-volume daily delivery routes with consistent accuracy, applying strong time management and situational judgment in fast-paced, unpredictable conditions.
- Used digital tracking systems to monitor route progress and update records, resolving delivery discrepancies independently against strict operational deadlines.

The UPS Store
Customer Service Representative

Ottawa, ON
Nov 2023 – Sept 2024

- Improved transaction accuracy by monitoring order processing systems and identifying discrepancies in package tracking workflows.
- Reduced new employee onboarding time by training staff on operational systems and internal technical procedures.
- Resolved customer shipment issues by troubleshooting system errors within internal tracking software, maintaining service efficiency.

VOLUNTEER EXPERIENCE

OrKidstra
WordPress Developer (Volunteer)

Ottawa, ON
April 2026 – Present

- Improve site accessibility and usability by auditing WCAG violations and redesigning page layouts, serving the public with accurate, clearly structured program information.
- Maintain and update public-facing web content, ensuring information integrity and consistent presentation that meets organizational and community standards.
- Document site changes and update internal records to support team handoffs and consistent content management practices.

EDUCATION

Algonquin College of Applied Arts and Technology
Diploma in Web Development and Internet Applications

Ottawa, ON
2025

PROJECTS

TrainerOS – Full-Stack Fitness Coaching Platform traineros.me

ASP.NET Core 8, C#, Entity Framework Core, PostgreSQL (Neon), React 19, TypeScript, Vite, Tailwind CSS, Azure App Service, Azure Functions, Queue Storage, Application Insights, GitHub Actions, Resend | Role: Solo Full-Stack Developer

- Designed and shipped a fitness coaching platform where trainers build per-client programs and clients log workouts on their phone between sets, live in production with a real client. Built solo from five frozen design documents through to automated GitHub Actions deployment to Azure App Service and Azure Functions.
- Built an asynchronous notification pipeline on Azure Functions and Queue Storage with at-least-once delivery, enforcing idempotency through a unique database index rather than application logic, with retry backoff verified at 62s and 302s against a 1/5/15/60 specification, poison-queue dead-lettering, and a pending-sweep recovery path that recovered a real enqueue failure during testing.
- Implemented DST-safe occurrence math with wall-clock semantics, resolving spring-forward gaps and fall-back ambiguity to the earlier instant, validated against real 2026 transitions across three timezones.
- Enforced tenant isolation structurally by restricting owned-entity DbSet to internal scope so scoped-query extensions are the only compiling data-access path from either host, verified by a reflection tripwire test, and returning uniform 404 responses so cross-tenant records cannot be probed.
- Adopted mutation testing as standard review practice, verifying that each test fails when its mechanism is broken; caught seven tests that passed while proving nothing, including one pinning a production bug as correct behaviour.

Plant Plotter – Full-Stack Garden Planning Application plantplotter.me

Next.js, React, Express.js, MySQL, JWT, Vercel, Azure App Service, Azure VM, Resend | Role: Full-Stack Developer

- Designed and deployed a full-stack web application following Agile SDLC practices, independently expanding a group project into a live portfolio product with a decoupled frontend, RESTful back-end API, and cloud-hosted MySQL database.
- Optimized back-end performance by reducing duplicate API calls, adding summary endpoints, caching plant library data, applying MySQL indexes, and upgrading Azure VM resources, demonstrating applied system design and data structure thinking.
- Implemented a secure authentication system including protected routes, expired-token handling, account deletion, and password reset with expiring hashed tokens, applying OOP principles and JWT-based authorization patterns.
- Deployed across Vercel (frontend), Azure App Service (back-end), and Azure VM (MySQL), including custom API domain, DNS configuration, CORS, environment variables, and production smoke testing.